

7 The ring-tailed lemur, *Lemur catta*, is a primate that is found only on the island of Madagascar.

Figure 7.1 shows a ring-tailed lemur.

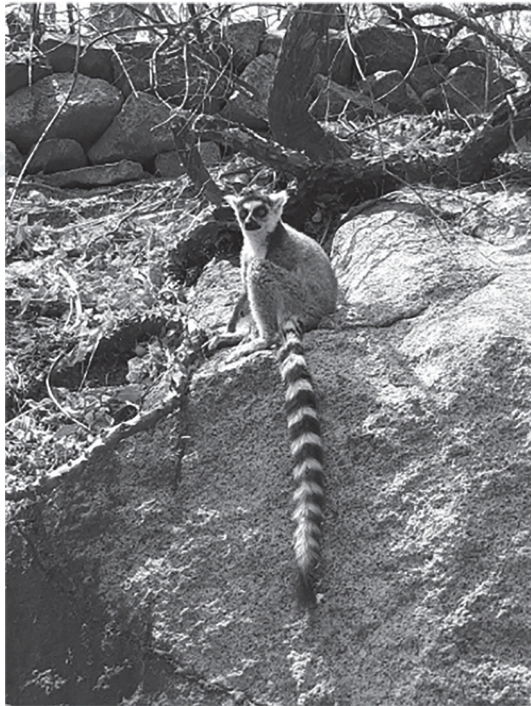


Figure 7.1

- (a)** Ring-tailed lemurs are classified as endangered on the International Union for Conservation of Nature (IUCN) Red List of Threatened Species™. There are approximately 2000 ring-tailed lemurs in the wild in Madagascar.

Suggest reasons why the ring-tailed lemur has become endangered.

[3]

- (b) Zoos have a role in protecting endangered species.

Outline the methods of assisted reproduction used by zoos in the conservation of endangered mammals such as the ring-tailed lemur.

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- (c) One important food source for the ring-tailed lemur is the hissing cockroach, *Gromphadorhina portentosa*. The hissing cockroach is a large, flightless insect that lives on forest floors in Madagascar.

- (i) The mark-release-recapture method was used to estimate the size of a population of the hissing cockroach.

- 33 cockroaches were trapped, marked and released.
- A few days later 46 cockroaches were trapped, of which 9 were marked.

The Lincoln index is used to estimate the size of a population.

The formula for the Lincoln index is:

$$N = \frac{n_1 \times n_2}{m_2}$$

Key

N = estimate of population size

n_1 = number of individuals captured in first sample

n_2 = number of individuals (both marked and unmarked) captured in second sample

m_2 = number of marked individuals recaptured in second sample



Use the Lincoln index to estimate the size of this population of hissing cockroaches.

Give your answer to the nearest whole number. Show your working.

estimate of size of population = [2]

- (ii) Suggest why the mark-release-recapture method is a suitable method to use when estimating the population size of the hissing cockroach.

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[Total: 12]